Environment and preferences

At date t, there are Y_t young and O_t old households. Each entering cohort has heterogeneous $(y_i^y, b_i, y_i^o) \sim F$; the young choose fertility and retain tenure when old.

$$x = c - \chi n > 0, \qquad s = h - \kappa n > 0,$$

$$u_t^y = \log x + \alpha \log s + \vartheta_t \log n, \qquad u^o = \log c^o + \gamma \log h^o + \omega_B \log e.$$

Physical housing caps are $h^d \leq h_d^{\max}$, with $h_R^{\max} < h_O^{\max}$; q is the bond price, P_t the house price, and τ_t^p the property-tax rate.

Household choices

Young households maximize $u_t^y + \beta V_{t+1}^d$ over c, h, n, a':

$$\begin{aligned}
&\text{Rent:} \quad c + qa' + u_t h = y_i^y + b_i + T_t, && a' \geq 0,\\
&\text{Own:} \quad c + qa' + (1 + q\tau_t^p) P_t h = y_i^y + b_i + T_t, && qa' + \phi_t P_t h \geq 0.
\end{aligned}$$

Old households maximize u^o; H is their inherited housing:

$$\begin{aligned}
c^o + qe + u_t h^o &= a + P_t H + y_i^o + T_t,\\
e &\geq P_{t+1} h^o \quad \text{for owners;} \qquad H = 0 \quad \text{for renters.}
\end{aligned}$$

Both ages respect their tenure's size limit. The young choose ownership if $W_t^O(i) + \xi_i \geq W_t^R(i)$, with logistic taste ξ_i.

Equilibrium

Rental competition and fiscal rebates satisfy

$$u_t = qr_t = (1 + q\tau_t^p)P_t - qP_{t+1}, \qquad (Y_t + O_t)T_t = q\tau_t^p P_t \bar{H}.$$

Housing and demography clear:

$$Y_t \bar{h}_t^y + O_t \bar{h}_t^o = \bar{H}, \qquad Y_{t+1} = \nu \bar{n}_t Y_t, \qquad O_{t+1} = Y_t.$$

A positive stationary equilibrium has

$$Y = O = N, \qquad \bar{n} = 1/\nu, \qquad N = \frac{\bar{H}}{\bar{h}^y + \bar{h}^o}.$$

The planner's allocation

Fix fertility, tenure, future opportunities and net estates. The planner chooses all current c, h and can relax individual financial constraints:

$$\max \int \left[\log(c_i^y - \chi n_i) + \alpha \log(h_i^y - \kappa n_i) + \log c_i^o + \gamma \log h_i^o\right] \mathrm{d}Q$$

$$\int (c_i^y + c_i^o)\mathrm{d}Q = C^{eq}/N, \qquad \int (h_i^y + h_i^o)\mathrm{d}Q = \bar{H}/N, \qquad h_i^y, h_i^o \leq H_i.$$

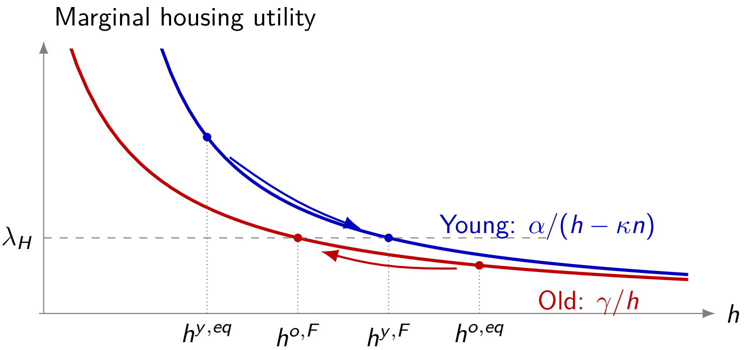
Q is the household distribution, H_i the tenure's size limit, and C^{eq} total reference consumption; both age groups have mass N.

A sufficient benchmark has $\phi = q$, zero tax, and slack competitive size and old financial-estate limits. Write $w_i = y_i^y + b_i$, $v_i = y_i^o$, $E = 1 + \alpha + \vartheta$, $K = 1 + \gamma + \omega_B$.

$$qEv_i > \beta K w_i \text{ for all } i, \qquad \frac{\bar{v}}{K} > \frac{\bar{w}}{E} \quad \Longrightarrow \quad H_Y^F > H_Y^{eq}.$$

Future income cannot be fully brought forward to finance young housing.

Housing allocation



Matched owners with fixed fertility and slack size limits. At the planner's allocation, the marginal utility of housing is equal across households.

Fertility

For a parent receiving a changed consumption–housing bundle,

$$\mathrm{d}n = \frac{\chi \,\mathrm{d}c/x^2 + \alpha\kappa \,\mathrm{d}h/s^2}{\vartheta/n^2 + \chi^2/x^2 + \alpha\kappa^2/s^2}.$$

More space raises fertility if the accompanying loss of goods is small enough.

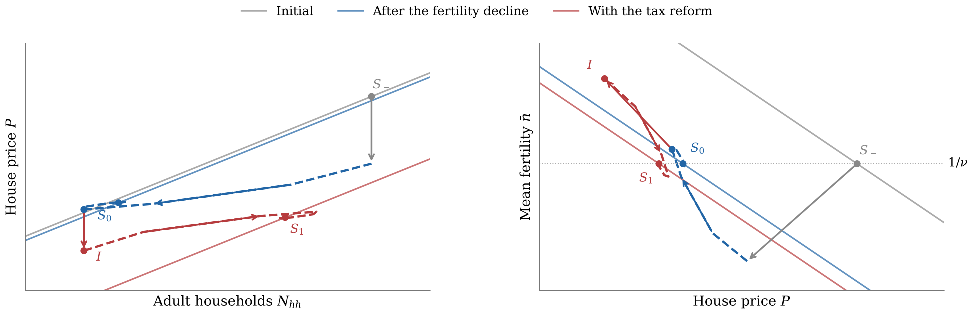
If the planner chooses fertility jointly with consumption and housing,

$$\bar{c}^{o,eq} \geq \bar{x}, \qquad \bar{h}^{o,eq} \geq \frac{\gamma}{\alpha}\bar{s}, \qquad h_R^{\max} > \bar{h}^{y,eq} \quad \Longrightarrow \quad \bar{n}^J > \bar{n},$$

with at least one of the first two comparisons strict. These conditions hold under the preceding income benchmark; planner caps may bind.

Average private fertility after a fixed-fertility allocation need not rise. The joint planner accounts for the adjustment of all resources together.

Demographic transition



Local equilibrium illustration. Solid: stationary market schedules. Dashed: first-order transition.

An exogenous fertility-taste decline starts the baseline transition. A permanent rebated tax introduced at I raises fertility on impact and leads to $N_1 > N_0$, under the note's local conditions. Both terminal fertility rates equal $1/\nu$.